

SHUBHAM KUMAR YADAV

AI-NATIVE FULL-STACK ENGINEER · LLM, RAG & AI AGENTS · CLOUD & DEVOPS

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coshubham.github.io/Portfolio- | github.com/coshubham | [LinkedIn](#)

Core skills: Python | Machine Learning & Deep Learning | LLM, VLM and Vector Search | FastAPI | REST API | React | TypeScript | PostgreSQL | OpenSearch | Docker | Azure | AWS | CI/CD

SUMMARY

AI-native full-stack engineer who designs, builds and runs **production AI systems end to end**, from the LLM, RAG, AI-agent and vector-search layer to the FastAPI and React layers on top and the cloud and DevOps underneath. Technical owner of **bCloud AI Search**, a multi-tenant AI product-search SaaS on Azure, and the engineer companies turn to when a live platform needs to be **restored, secured or taken over**. I **lead a small development team**, work directly with clients, and ship work that stays reliable in production. Open to relocation and remote roles.

EXPERIENCE

ATH Infosystems, Noida — Full Stack Engineer (AI/ML, DevOps & Cloud)

Present

- Delivered **11+ projects to production** for ATH Infosystems and its clients across AI search, e-commerce, analytics, FinTech and identity automation, owning each from planning to release.
- Technical owner of **bCloud AI Search**, a multi-tenant AI product-search SaaS on Azure: **70+ FastAPI endpoints**, OpenAI embeddings with OpenSearch k-NN, two React dashboards and the Docker, nginx and TLS stack; cut search latency from **~500 ms to ~150-250 ms**.
- Shipped **voice, conversational, multi-product and photo search**, an **AI helpdesk agent** on OpenAI tool calling, and a **31-test tenant-isolation suite** that keeps every customer's data separate.
- Restored a large e-commerce marketplace's dead AWS search platform on a frozen backend, re-indexing **255,350 products with zero errors** in 1 h 52 min; built a nightly ML risk model at **0.936 ROC AUC** with **55+ DAX measures** in Power BI and Microsoft Fabric.
- Lead a small development team** and own client communication from requirements to support; took over **SplitIT**, a live FinTech platform, ran its security audit and the client-approved production pause, and wrote a **51-page technical handbook** and deployment runbooks.

Salesqueen Software Solutions, Chennai — Front-End Developer (Internship)

Mar 2025 – Jul 2025

Remote, alongside final-semester MCA

- Built **responsive, accessible UI components** across **six delivered products**, including a login portal, production portal, Kanban board, e-commerce website and an online cab-booking app.
- Deployed and maintained releases on live servers** and kept the interface layer consistent across the product suite.

CSRBOX Foundation via IBM SkillsBuild — AI Intern (Remote)

Jul 2024 – Aug 2024

- Built a **conversational chatbot** using **IBM Watsonx AI** and NLP techniques.

Gautam Buddha University, Greater Noida — Admission Department Intern

Jun 2024 – Aug 2024

- Managed a **3,000+ student record dataset in MySQL** plus CRM data through the admission cycle, and built an **online Bonafide Certificate system** that issued certificates from those records.

EDUCATION

Gautam Buddha University — MCA (Artificial Intelligence), CGPA 8.71

Passed

Veer Narmad South Gujarat University — BCA, CGPA 7.61

Passed

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript (ES6+), TypeScript, SQL, DAX, Bash, PHP, Kotlin, HTML5, CSS3
- **AI / Machine Learning:** machine learning and **deep learning**, **PyTorch**, scikit-learn, **Vision-Language Models (VLM)**, **Vision-Language-Action (VLA) models** ($\pi 0$ / OpenPI, LeRobot, LIBERO), **imitation learning**, **diffusion and flow-matching policies**, **LoRA fine-tuning**, model evaluation, computer vision and image search, speech-to-text and text-to-speech, NLP, PySpark, IBM Watsonx AI
- **LLM, RAG & Search:** OpenAI text and **multimodal embeddings**, **LLM tool-calling and AI agents**, prompt & tool design, **RAG (Retrieval-Augmented Generation)**, semantic and **vector search (k-NN)**, OpenSearch (Elasticsearch-compatible), search relevance tuning, intent detection
- **Backend & APIs:** FastAPI, Flask, Node.js, Express.js, Python asyncio, REST API design, OpenAPI / Swagger, GraphQL, JWT, OAuth2, webhooks, multi-tenant API architecture, Stripe API, PayPal API, BigCommerce API, Microsoft Graph API, SCIM
- **Frontend & Mobile:** React, Vite, Tailwind CSS, Bootstrap 5, Recharts, Chart.js, Three.js / WebGL, Web Speech API, responsive and accessible UI, Android (Kotlin, Firebase)
- **Cloud & DevOps:** Azure (VMs, Logic Apps, Key Vault, Entra ID, DevOps), AWS (EC2, ECS Fargate, Lambda, SQS, RDS, S3, CloudFront, IAM, OpenSearch Serverless, ElastiCache, Secrets Manager), Terraform, Docker, Docker Compose, Ansible, GitHub Actions, CI/CD, nginx, TLS/certbot, Linux (Ubuntu, cron)
- **Databases:** PostgreSQL, MySQL, Microsoft SQL Server, MongoDB (Atlas), Redis, SQLite, Firebase
- **BI & Data:** Microsoft Fabric, Power BI, DAX, Tabular Editor, Delta Lake / OneLake, Copilot Studio
- **Tools & Practices:** Git, GitHub, Postman, Figma, technical writing & runbooks, team leadership, code review, mentoring

CERTIFICATIONS

- DevOps and Cloud Computing — PW Skills · [Certificate](#)
- AI Internship, Watsonx AI — IBM SkillsBuild / CSRBOX · [Certificate](#)
- Front-End Developer — Salesqueen · [Certificate](#)
- Internship — GBU Admission Department · [Certificate](#)

PERSONAL PROJECTS

- **Portfolio website** ([Live](#) · [Code](#)) — 3D personal site in HTML, CSS, JavaScript and Three.js with case studies and a contact chooser, hosted on GitHub Pages.
- **CI/CD To-Do App** ([GitHub](#)) — React + Express app deployed to 11 AWS EC2 instances by Ansible, triggered from GitHub Actions.
- **Expense Tracker** ([Live Demo](#)) — React front end, GraphQL (Apollo) API, MongoDB and Passport login, live on Render.
- **Android Chat App** ([GitHub](#)) — Kotlin app on Firebase Auth and Realtime Database, with a web admin panel.
- **Watsonx Restaurant Chatbot** ([Live](#) · [Code](#)) — Conversational restaurant chatbot built on IBM Watsonx AI with NLP during the IBM SkillsBuild internship; runs in the browser.
- **FoodHut Online Ordering** ([Source](#)) — Restaurant food-ordering website where customers browse the menu and place orders; PHP, MySQL, Bootstrap and JavaScript.

PRODUCTION PROJECTS

► bCloud AI Search — Multi-Tenant AI Product-Search SaaS

ATH Infosystems

bcloud.ai · ATH's own product · live in production · technical owner: architecture, planning, build and operations

Planned, built and now run ATH's multi-tenant AI search for online stores: OpenAI text-embedding-3-large (3072-dim) vectors with OpenSearch k-NN behind **70+ FastAPI endpoints**, and **five ways to search** — typed; voice with a spoken LLM answer; conversational queries through an LLM intent step that keeps price and stock limits; multi-product queries such as "laptop and headphones"; and photo search with a vision model plus image fingerprints. Also BigCommerce catalogue sync, a 5-minute trending-score job, an AI helpdesk agent, and **122 automated tests**.

► bCloud Admin Console & Client Portal

ATH Infosystems

React · Vite · Tailwind CSS · Recharts · the two dashboards of bCloud AI Search

A full admin console (clients, field mapping, OpenSearch, Azure, AI, monitoring, billing, support tickets, search analytics) and a light/dark client portal where each store tracks its search analytics, sync runs, billing with a **month-end forecast**, and "missed searches" with **one-click synonym fixes** checked live against its own catalogue.

► Legacy Marketplace Search Platform — Restore & Modernisation

Large e-commerce marketplace client · ATH Infosystems

Engineered a local TLS/SigV4 compatibility shim so code written for AWS OpenSearch Serverless ran unmodified against self-hosted OpenSearch (**zero changes to the client's code**). Built a FastAPI service powering two new React dashboards behind nginx, wrote idempotent bash installers that **redploy the stack in one command**, and fixed a silent analytics event-loss bug.

► Predictive Analytics & Reporting Platform — Power BI and Microsoft Fabric

Education-sector client · ATH Infosystems

Enrolment forecasting, retention and admissions analytics: layered semantic models, cross-source CRM-to-SIS integration with **row-level security**, and an **AI assistant** answering plain-English questions from governed DAX measures.

► AI Lead-Capture Assistant

ATH Infosystems

Built a Flask chat service and embeddable JavaScript widget live on the company's two public sites: answers visitor questions through an LLM and **captures qualified leads into SQL Server**, with a lead admin console, CSV export and Swagger-documented APIs. Authored **44 of the project's 60 commits**.

► SplitIT — FinTech Income-Splitting SaaS

FinTech client · live production · inherited ownership · ATH Infosystems

Took over a live platform that splits a user's incoming income across payees by rules and pays them out through Stripe Connect and ACH debits, with PayPal Billing subscriptions — React, TypeScript, Express and PostgreSQL on AWS (ECS Fargate, RDS, CloudFront). Ran a **security audit** (unauthenticated payment endpoint, IDOR, sensitive data in logs, committed secrets) with a remediation plan; **found the missing PayPal webhook** that let subscriptions expire while customers kept being billed; then executed the client-approved production pause — full backups to S3, database kept as snapshots, service and load balancer shut down — and wrote the handover and relaunch guide.

► Automated HR-to-Cloud Identity Provisioning Pipeline

Corporate client · HR and identity automation · ATH Infosystems

Azure Logic Apps pipeline syncing BambooHR to Microsoft Entra ID via the Microsoft Graph API and SCIM bulk upload, with OAuth2, Azure Key Vault, duplicate-conflict detection, automated group assignment and Office 365 welcome emails — a self-healing idempotent design that **removed manual onboarding entirely**.